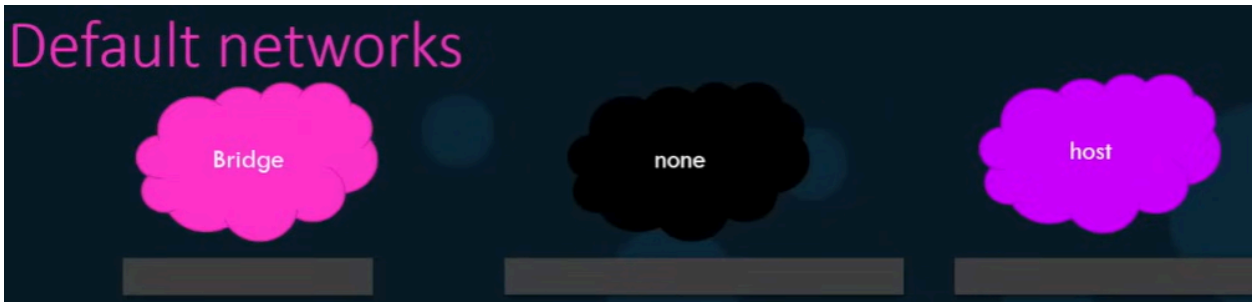
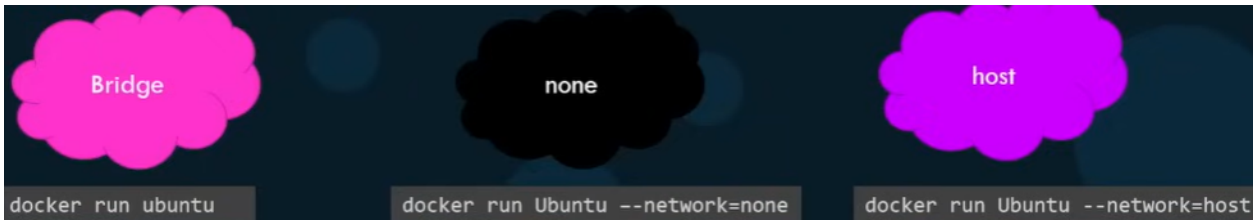
Docker Networking

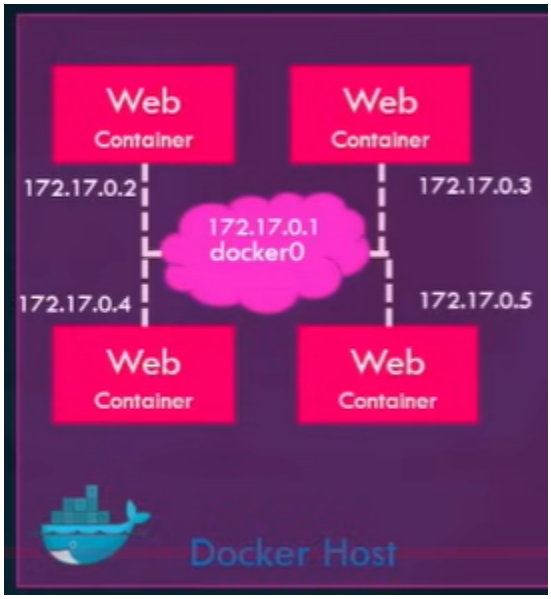
After installing docker, it install three networks automatically, **Bridge, none and host.**



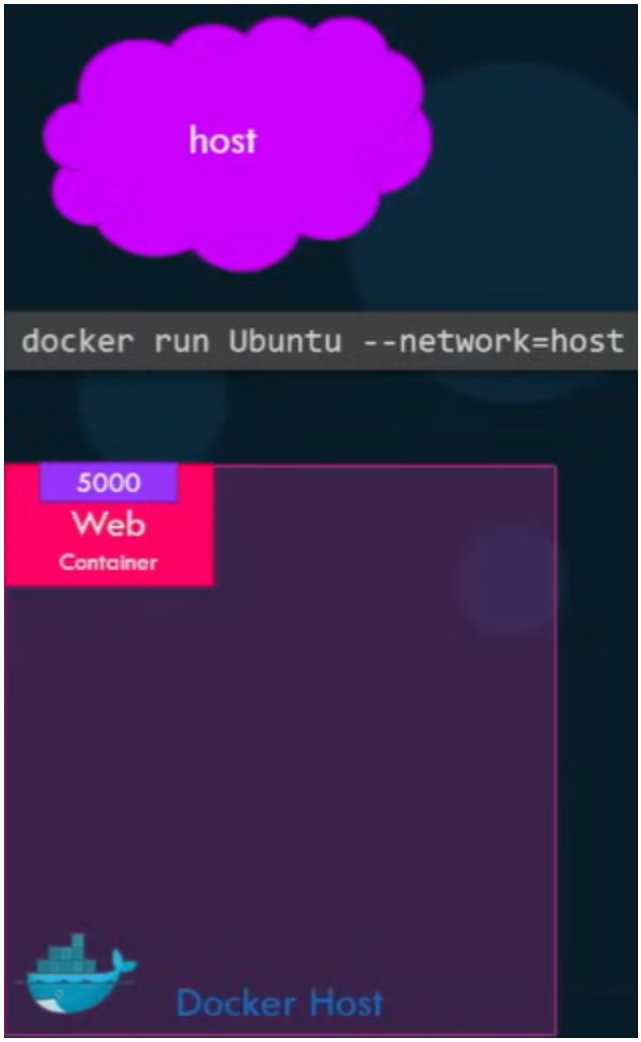
Bridge: a default network that container gets attached to.
Bridge: **docker run ubuntu**
None: **docker run Ubuntu –network=none**
Host: **docker run Ubuntu –network=host**



Bridge is the private internal network, created by docker on the host, all containers attached to this network by default, they get the internal ip address in range 172.17 series, the container can access each other using internal ip, if required,

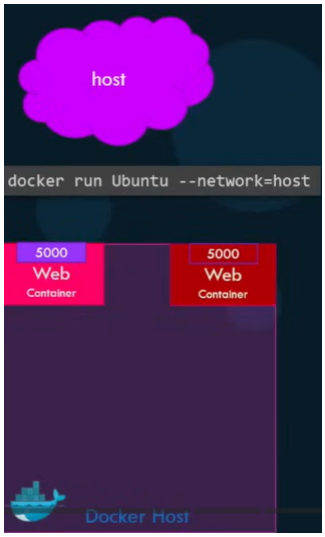


To access any of the container from outside, map the port, to the docker host.

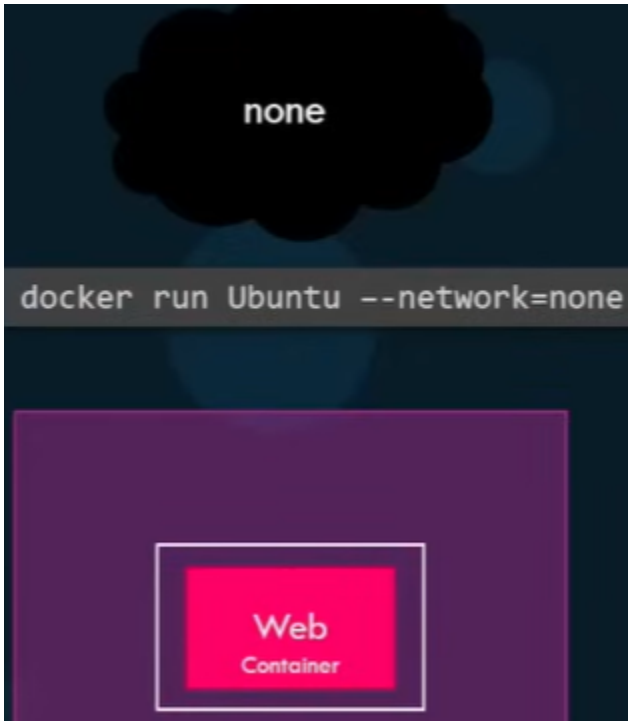


Another way of accessing containers externally, associate the container to host network, no need of port mapping as the web container uses the host

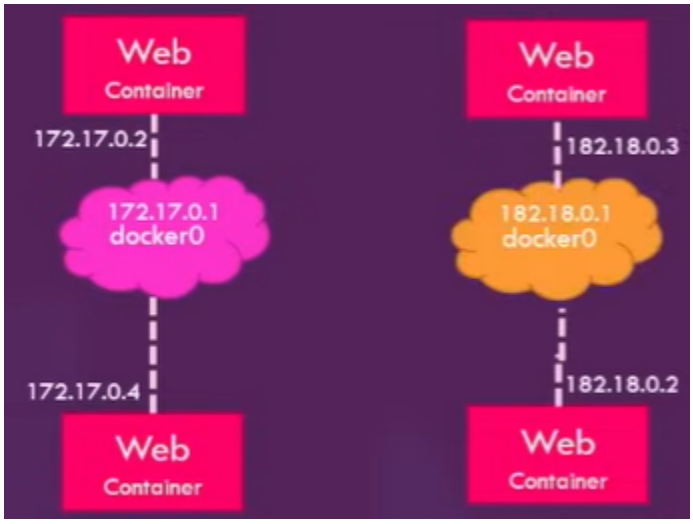
Unlike before, you Cannot run multiple containers on same host and same port.



For none network, the containers are not attached to any network. Does not have access to external network or other containers, they run in the isolated network.



For the default bridge network, Making it having internal networks as 172 and 182, like



by default docker create only one internal bridge network. We could create our own internal network using command

```
docker network create \
    --driver bridge \
    --subnet 182.18.0.0/16
    custom-isolated-network
```

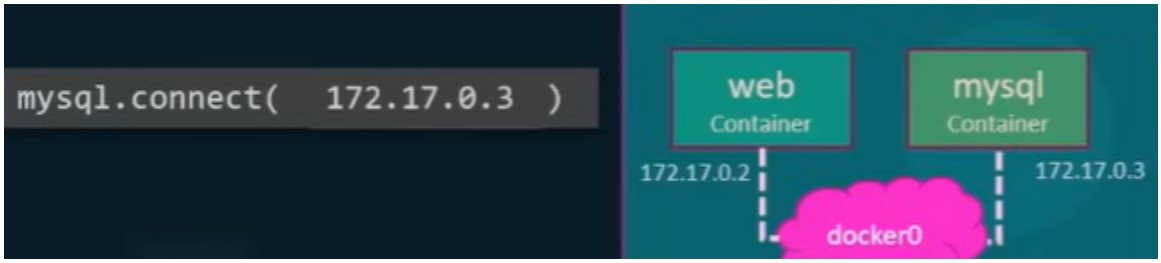
To list all networks:

```
docker network ls

root@osboxes:/root # docker network ls
NETWORK ID          NAME                        DRIVER              SCOPE
dba0fb9370fe        bridge                      bridge              local
46d476b87cd9        customer-isolated-network   bridge              local
6de685cec1ce        docker_gwbridge             bridge              local
e29d188b4e47        host                        host                local
mmrho7vsb9rm        ingress                     overlay             swarm
d9f11695f0d6        none                        null                local
d371b4009142        simplewebappdocker_default  bridge              local
```

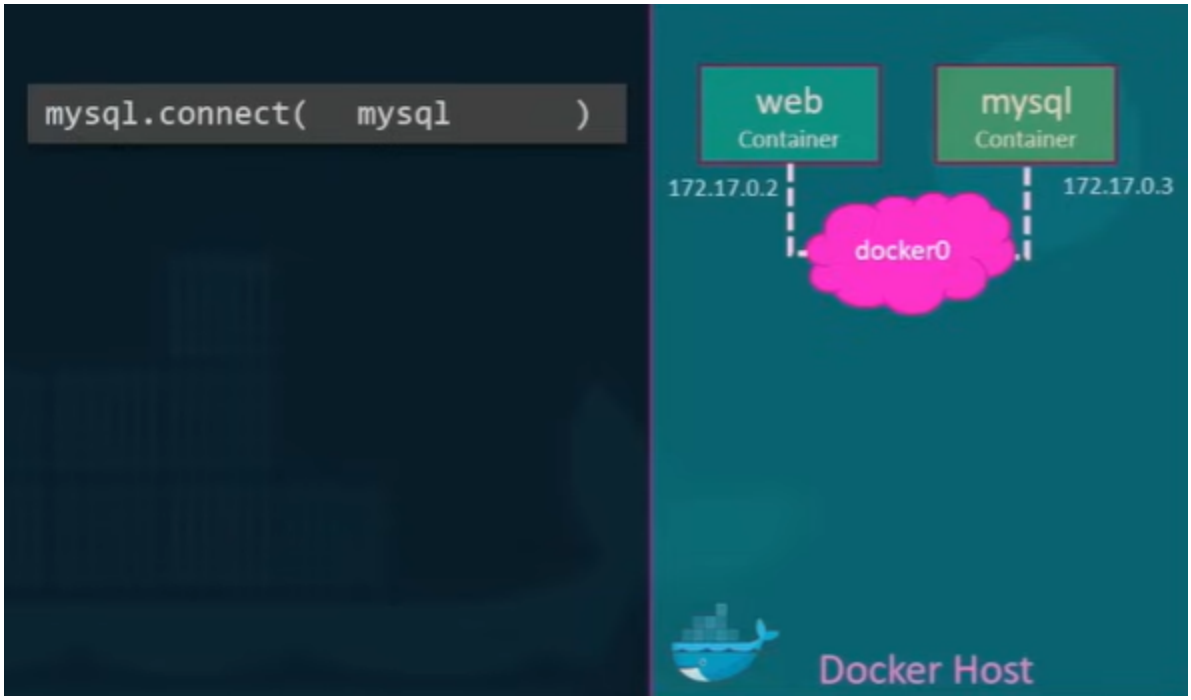
To inspect the container,

```
docker inspect blissful_hopper

[
    {
        "Id": "35505f7810d17291261a43391d4b6c0846594d41
        "Name": "/blissful_hopper",
        "NetworkSettings": {
            "Bridge": "",
            "Gateway": "172.17.0.1",
            "IPAddress": "172.17.0.6",
            "MacAddress": "02:42:ac:11:00:06",
            "Networks": {
                "bridge": {
                    "Gateway": "172.17.0.1",
                    "IPAddress": "172.17.0.6",
                    "MacAddress": "02:42:ac:11:00:06",
                }
            }
        }
]
```

Containers can reach each other using their names, Example: web server and mysql database containers are running in the same node, how can a webserver can access the database container, one way is by using internal ip address of the database container



Ip address wll not be same on the system reboot. Right way to do it is by using container name.all container in the docker host can resolve each other with the name of the container.



Docker has a built-in DNS server, that helps the containers to resolve each other using the container name. Note that built-in DNs server runs at 127.0.0.11. So how does docker implement networking, what's the technology behind it, how were the containers isolated within the host?

Docker uses network name spaces, that creates the separate name space for each container, and uses virtual ethernet pairs, to connect containers together.